

Untitled10

June 5, 2025

1 AULA SPARK

```
[2]: from pyspark.sql import SparkSession
```

```
[3]: spark = SparkSession.builder.appName('Aula de Spark').enableHiveSupport().  
      ↪getOrCreate()
```

```
[4]: spark
```

```
[4]: <pyspark.sql.session.SparkSession at 0x7fe064ca5cf0>
```

```
[ ]: #ler dados do hdfs
```

```
[12]: df_hdfs = spark.read.csv(path='/datalake/arquivo/',header=True)
```

```
[13]: df_hdfs.show()
```

```
+-----+-----+-----+  
|usuario|item|avaliacao|  
+-----+-----+-----+  
|    109|   9|         3|  
|    174| 412|         1|  
|     7| 208|         5|  
|   371|  97|         5|  
|   296| 255|         2|  
|   280|  82|         2|  
|   271| 275|         4|  
|   110| 791|         2|  
|    59| 926|         1|  
|   217| 576|         1|  
|   145| 665|         5|  
|   334| 204|         4|  
|    42| 568|         4|  
|   200| 143|         5|  
|    89| 387|         5|  
|   311| 588|         4|  
|   235| 269|         4|  
|   287| 156|         5|
```

```
|      344| 204|      4|
|      43| 289|      4|
+-----+-----+-----+
only showing top 20 rows
```

```
[ ]: #ler dados do s3
```

```
[14]: df_s3 = spark.read.json('s3a://camada-bronze/city')
```

```
[15]: df_s3.show()
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+
|__deleted|  __lsn|__op|__source_ts_ms|__table|      city|city_id|country_id|
last_update|
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+
|      false|92662824|  c| 1678318110082|  city|Divinolandia|  652|
11|1678307310059665|
|      false|92662112|  c| 1678317762797|  city|  Araraquara|  651|
11|1678306962794627|
|      false|92685840|  c| 1678318113380|  city|    New York|  653|
11|1678307313380279|
|      false|92639248|  c| 1678317758356|  city|      Santos|  650|
11|1678306958332527|
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+
```

```
[16]: #ler dados de banco de dados
```

```
[17]: conn = 'jdbc:postgresql://postgres:5432/dvdrental'
```

```
[18]: prop = {
      "user": "admin",
      "password": "admin",
      "driver": "org.postgresql.Driver"
    }
```

```
[20]: df_city = spark.read.jdbc(url=conn,properties=prop,table='public.city')
```

```
[23]: df_city.count()
```

```
[23]: 652
```

```
[28]: query = '(select c.customer_id,c.first_name, c.email ,c2.city from public.
↳customer c \
inner join public.address a on c.address_id = a.address_id \
inner join public.city c2 on c2.city_id = a.city_id) as tab'
```

```
[29]: query
```

```
[29]: '(select c.customer_id,c.first_name, c.email ,c2.city from public.customer c
inner join public.address a on c.address_id = a.address_id inner join
public.city c2 on c2.city_id = a.city_id) as tab'
```

```
[30]: df_query = spark.read.jdbc(url=conn,properties=prop, table=query)
```

```
[32]: df_query.show(truncate=False)
```

```
+-----+-----+-----+-----+
|customer_id|first_name|email                                |city          |
+-----+-----+-----+-----+
|524        |Jared     |jared.ely@sakilacustomer.org        |Purwakarta    |
|1          |Mary      |mary.smith@sakilacustomer.org       |Sasebo        |
|2          |Patricia  |patricia.johnson@sakilacustomer.org |San Bernardino|
|3          |Linda     |linda.williams@sakilacustomer.org   |Athenai       |
|4          |Barbara   |barbara.jones@sakilacustomer.org    |Myingyan      |
|5          |Elizabeth |elizabeth.brown@sakilacustomer.org  |Nantou        |
|6          |Jennifer  |jennifer.davis@sakilacustomer.org   |Laredo        |
|7          |Maria     |maria.miller@sakilacustomer.org     |Kragujevac    |
|8          |Susan     |susan.wilson@sakilacustomer.org     |Hamilton      |
|9          |Margaret  |margaret.moore@sakilacustomer.org   |Masqat        |
|10         |Dorothy   |dorothy.taylor@sakilacustomer.org   |Esfahan       |
|11         |Lisa      |lisa.anderson@sakilacustomer.org    |Sagamihara    |
|12         |Nancy     |nancy.thomas@sakilacustomer.org     |Yamuna Nagar  |
|13         |Karen     |karen.jackson@sakilacustomer.org    |Osmaniye      |
|14         |Betty     |betty.white@sakilacustomer.org      |Citrus Heights|
|15         |Helen     |helen.harris@sakilacustomer.org     |Bhopal        |
|16         |Sandra    |sandra.martin@sakilacustomer.org    |Southend-on-Sea|
|17         |Donna     |donna.thompson@sakilacustomer.org   |Elista        |
|18         |Carol     |carol.garcia@sakilacustomer.org     |Kaduna        |
|19         |Ruth      |ruth.martinez@sakilacustomer.org    |Kimberley     |
+-----+-----+-----+-----+
```

only showing top 20 rows

```
[33]: #salvo os dados
```

```
[36]: df_query.show(2)
```

```
+-----+-----+-----+-----+
```

customer_id	first_name	email	city
524	Jared	jared.ely@sakilac...	Purwakarta
1	Mary	mary.smith@sakila...	Sasebo

only showing top 2 rows

```
[37]: type(df_query)
```

```
[37]: pyspark.sql.dataframe.DataFrame
```

```
[41]: df_query.write.csv('hdfs:///datalake/query',header=True,sep=';')
```

```
-----
AnalysisException                                Traceback (most recent call last)
Cell In[41], line 1
----> 1 df_query.write.csv('hdfs:///datalake/query',header=True,sep=';')

File /usr/local/spark/python/pyspark/sql/readwriter.py:1240, in DataFrameWriter
    csv(self, path, mode, compression, sep, quote, escape, header, nullValue,
    escapeQuotes, quoteAll, dateFormat, timestampFormat, ignoreLeadingWhiteSpace,
    ignoreTrailingWhiteSpace, charToEscapeQuoteEscaping, encoding, emptyValue,
    lineSep)
    1221 self.mode(mode)
    1222 self._set_opts(
    1223     compression=compression,
    1224     sep=sep,
    (...)
    1238     lineSep=lineSep,
    1239 )
-> 1240 self._jwrite.csv(path)

File /usr/local/spark/python/lib/py4j-0.10.9.5-src.zip/py4j/java_gateway.py:
    1321, in JavaMember.__call__(self, *args)
    1315 command = proto.CALL_COMMAND_NAME + \
    1316     self.command_header + \
    1317     args_command + \
    1318     proto.END_COMMAND_PART
    1320 answer = self.gateway_client.send_command(command)
-> 1321 return_value = get_return_value(
    1322     answer, self.gateway_client, self.target_id, self.name)
    1324 for temp_arg in temp_args:
    1325     temp_arg._detach()

File /usr/local/spark/python/pyspark/sql/utils.py:196, in capture_sql_exception
    192 converted = convert_exception(e.java_exception)
```

```

193 if not isinstance(converted, UnknownException):
194     # Hide where the exception came from that shows a non-Pythonic
195     # JVM exception message.
--> 196     raise converted from None
197 else:
198     raise

```

`AnalysisException`: path hdfs://namenode:8020/datalake/query already exists.

```
[40]: df_query = df_query.repartition(3)
```

```
[42]: df_query.write.csv('hdfs:///datalake/query',header=True,sep=';
    ↪',mode='overwrite')
```

```
[43]: df_query.write.json('s3a://camada-bronze/query')
```

```
[44]: df_query.write.parquet('s3a://camada-prata/query')
```

```
[45]: df_query.write.format('hive').saveAsTable('default.query')
```

```
[49]: df_query.write.mode('append').format('hive').insertInto('default.query')
```

```
[50]: #ler dados do hive
```

```
[51]: df_hive = spark.read.table('default.query')
```

```
[52]: df_hive.show()
```

```

+-----+-----+-----+-----+
|customer_id|first_name|email|city|
+-----+-----+-----+-----+
|366|Brandon|brandon.huey@saki...|Balikesir|
|138|Hazel|hazel.warren@saki...|Hohhot|
|255|Irma|irma.pearson@saki...|Hagonoy|
|536|Fernando|fernando.churchil...|Tonghae|
|520|Mitchell|mitchell.westmore...|Nha Trang|
|238|Nellie|nellie.garrett@sa...|Shimoga|
|183|Ida|ida.andrews@sakil...|Luzinia|
|438|Barry|barry.lovelace@sa...|Kitwe|
|571|Johnnie|johnnie.chisholm@...|Plock|
|579|Daryl|daryl.larue@sakil...|Mosul|
|213|Gina|gina.williamson@s...|Taizz|
|92|Tina|tina.simmons@saki...|Goinia|
|356|Gerald|gerald.fultz@saki...|Satna|
|73|Beverly|beverly.brooks@sa...|Chiayi|
|43|Christine|christine.roberts...|Faaa|
|194|Kristen|kristen.chavez@sa...|Hino|

```

```

|      435|      Ricky|ricky.shelby@saki...|      Junan|
|      134|      Emma|emma.boyd@sakilac...|      Qalyub|
|      552|      Hugh|hugh.waldrop@saki...|      Haining|
|      206|      Terri|terri.vasquez@sak...|      Blumenau|
+-----+-----+-----+-----+
only showing top 20 rows

```

```
[53]: df_hive.join(df_query,df_query.customer_id == df_hive.customer_id).show()
```

```

+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|customer_id|first_name|email|city|customer_id|first_name|email|city|
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|      559|      Everett|everett.banda@sak...|      Bilbays|      559|
Everett|everett.banda@sak...|      Bilbays|
|      449|      Oscar|oscar.aquino@saki...|      Sirjan|      449|
Oscar|oscar.aquino@saki...|      Sirjan|
|      250|      Jo|jo.fowler@sakilac...|      Oyo|      250|
Jo|jo.fowler@sakilac...|      Oyo|
|      85|      Anne|anne.powell@sakil...|      Bradford|      85|
Anne|anne.powell@sakil...|      Bradford|
|      113|      Cindy|cindy.fisher@saki...|      Cuman|      113|
Cindy|cindy.fisher@saki...|      Cuman|
|      293|      Mae|mae.fletcher@saki...|Donostia-San Seba...|      293|
Mae|mae.fletcher@saki...|Donostia-San Seba...|
|      358|      Samuel|samuel.marlow@sak...|      Ranchi|      358|
Samuel|samuel.marlow@sak...|      Ranchi|
|      157|      Darlene|darlene.rose@saki...|      Pyongyang|      157|
Darlene|darlene.rose@saki...|      Pyongyang|
|      180|      Stacy|stacy.cunningham@...|      Pereira|      180|
Stacy|stacy.cunningham@...|      Pereira|
|      509|      Raul|raul.fortier@saki...|      Chapra|      509|
Raul|raul.fortier@saki...|      Chapra|
|      77|      Jane|jane.bennett@saki...|      Araatuba|      77|
Jane|jane.bennett@saki...|      Araatuba|
|      349|      Joe|joe.gilliland@sak...|      Imus|      349|
Joe|joe.gilliland@sak...|      Imus|
|      523|      Harvey|harvey.guajardo@s...|      Vinh|      523|
Harvey|harvey.guajardo@s...|      Vinh|
|      128|      Marjorie|marjorie.tucker@s...|      Jastrzebie-Zdrj|      128|
Marjorie|marjorie.tucker@s...|      Jastrzebie-Zdrj|
|      371|      Billy|billy.poulin@saki...|      Cabuyao|      371|
Billy|billy.poulin@saki...|      Cabuyao|
|      70|      Christina|christina.ramirez...|      al-Hawiya|      70|
Christina|christina.ramirez...|      al-Hawiya|

```

592	Terrance	terrance.roush@sa...	Szkesfehrvr	592
Terrance	terrance.roush@sa...	Szkesfehrvr		
544	Cody	cody.nolen@sakila...	So Bernardo do Campo	544
Cody	cody.nolen@sakila...	So Bernardo do Campo		
211	Stacey	stacey.montgomery...	Fuyu	211
Stacey	stacey.montgomery...	Fuyu		
399	Danny	danny.isom@sakila...	Okara	399
Danny	danny.isom@sakila...	Okara		

only showing top 20 rows

```
[54]: #ler dados com SQL
```

```
[60]: df_total = spark.sql('select * from parquet.`s3a://camada-prata/query`')
```

```
[61]: df_total.show()
```

customer_id	first_name	email	city
366	Brandon	brandon.huey@saki...	Balikesir
138	Hazel	hazel.warren@saki...	Hohhot
255	Irma	irma.pearson@saki...	Hagonoy
536	Fernando	fernando.churchil...	Tonghae
520	Mitchell	mitchell.westmore...	Nha Trang
238	Nellie	nellie.garrett@sa...	Shimoga
183	Ida	ida.andrews@sakil...	Luzinia
438	Barry	barry.lovelace@sa...	Kitwe
571	Johnnie	johnnie.chisholm@...	Plock
579	Daryl	daryl.larue@sakil...	Mosul
213	Gina	gina.williamson@s...	Taizz
92	Tina	tina.simmons@saki...	Goinia
356	Gerald	gerald.fultz@saki...	Satna
73	Beverly	beverly.brooks@sa...	Chiayi
43	Christine	christine.roberts...	Faaa
194	Kristen	kristen.chavez@sa...	Hino
435	Ricky	ricky.shelby@saki...	Junan
134	Emma	emma.boyd@sakilac...	Qalyub
552	Hugh	hugh.waldrop@saki...	Haining
206	Terri	terri.vasquez@sak...	Blumenau

only showing top 20 rows

```
[62]: #escrever no banco de dados
df_total.write.jdbc(url=conn, properties=prop, table='public.aula')
```

```
[63]: #api
```

```
[64]: import requests
```

```
[84]: l = []
      for x in range (10):
          print(x)
          retorno = requests.get('https://catfact.ninja/fact')
          l.append(retorno.json())
```

0
1
2
3
4
5
6
7
8
9

```
[85]: 1
```

```
[85]: [{'fact': 'Foods that should not be given to cats include onions, garlic, green
tomatoes, raw potatoes, chocolate, grapes, and raisins. Though milk is not
toxic, it can cause an upset stomach and gas. Tylenol and aspirin are extremely
toxic to cats, as are many common houseplants. Feeding cats dog food or canned
tuna that's for human consumption can cause malnutrition.',
      'length': 360},
      {'fact': 'The first formal cat show was held in England in 1871; in America, in
1895.',
      'length': 75},
      {'fact': 'Spanish-Jewish folklore recounts that Adam's first wife, Lilith,
became a black vampire cat, sucking the blood from sleeping babies. This may be
the root of the superstition that a cat will smother a sleeping baby or suck out
the child's breath.',
      'length': 245},
      {'fact': 'Kittens who are taken along on short, trouble-free car trips to town
tend to make good passengers when they get older. They get used to the sounds
and motions of traveling and make less connection between the car and the visits
to the vet.',
      'length': 239},
      {'fact': 'Cats should not be fed tuna exclusively, as it lacks taurine, an
essential nutrient required for good feline health. Make sure you have the
```



```
proper Pet supplies to keep your cat happy and healthy.',
  'length': 196},
  {'fact': 'Spanish-Jewish folklore recounts that Adam's first wife, Lilith,
became a black vampire cat, sucking the blood from sleeping babies. This may be
the root of the superstition that a cat will smother a sleeping baby or suck out
the child's breath.',
  'length': 245},
  {'fact': 'The smallest wildcat today is the Black-footed cat. The females are
less than 20 inches (50 cm) long and can weigh as little as 2.5 lbs (1.2 kg).',
  'length': 145},
  {'fact': 'The cat appears to be the only domestic companion animal not
mentioned in the Bible.',
  'length': 84},
  {'fact': 'A cat named Dusty, aged 17, living in Bonham, Texas, USA, gave birth
to her 420th kitten on June 23, 1952.',
  'length': 107},
  {'fact': "Cats must have fat in their diet because they can't produce it on
their own.",
  'length': 76}]
```

```
[67]: retorno.status_code
```

```
[67]: 200
```

```
[70]:
```

```
[70]: {'fact': 'The term "puss" is the root of the principal word for "cat" in the
Romanian term pisica and the root of secondary words in Lithuanian (puz) and Low
German\xa0puus. Some scholars suggest that "puss" could be imitative of the
hissing sound used to get a cat's attention. As a slang word for the female
pudenda, it could be associated with the connotation of a cat being soft, warm,
and fuzzy.',
  'length': 387}
```

```
[78]:
```

```
[86]: req = spark.read.json(spark.sparkContext.parallelize(1))
```

```
[88]: req.show()
```

```
+-----+-----+
|                fact|length|
+-----+-----+
|Foods that should...|   360|
|The first formal ...|    75|
|Spanish-Jewish fo...|   245|
|Kittens who are t...|   239|
```

```
|Cats should not b...| 196|
|Spanish-Jewish fo...| 245|
|The smallest wild...| 145|
|The cat appears t...| 84|
|A cat named Dusty...| 107|
|Cats must have fa...| 76|
+-----+-----+
```

```
[ ]: req.repartition(1).write.json('s3a://camada-bronze/facts',mode='append')
```

```
[94]: df_teste1 = spark.read.parquet('/datalake/query/*.parquet').show(1000)
```

```
+-----+-----+-----+-----+
|customer_id|first_name|email|city|
+-----+-----+-----+-----+
|559|Everett|everett.banda@sak...|Bilbays|
|449|Oscar|oscar.aquino@saki...|Sirjan|
|250|Jo|jo.fowler@sakilac...|Oyo|
|85|Anne|anne.powell@sakil...|Bradford|
|113|Cindy|cindy.fisher@saki...|Cuman|
|293|Mae|mae.fletcher@saki...|Donostia-San Seba...|
|358|Samuel|samuel.marlow@sak...|Ranchi|
|157|Darlene|darlene.rose@saki...|Pyongyang|
|180|Stacy|stacy.cunningham@...|Pereira|
|509|Raul|raul.fortier@saki...|Chapra|
|77|Jane|jane.bennett@saki...|Araatuba|
|349|Joe|joe.gilliland@sak...|Imus|
|523|Harvey|harvey.guajardo@s...|Vinh|
|128|Marjorie|marjorie.tucker@s...|Jastrzebie-Zdrj|
|371|Billy|billy.poulin@saki...|Cabuyao|
|70|Christina|christina.ramirez...|al-Hawiya|
|592|Terrance|terrance.roush@sa...|Szkesfehrvr|
|544|Cody|cody.nolen@sakila...|So Bernardo do Campo|
|211|Stacey|stacey.montgomery...|Fuyu|
|399|Danny|danny.isom@sakila...|Okara|
|266|Nora|nora.herrera@saki...|Salzburg|
|326|Jose|jose.andrew@sakil...|Kingstown|
|296|Ramona|ramona.hale@sakil...|Patiala|
|247|Stella|stella.moreno@sak...|Coacalco de Berri...|
|40|Amanda|amanda.carter@sak...|Nador|
|351|Jack|jack.foust@sakila...|Zeleznogorsk|
|99|Emily|emily.diaz@sakila...|Kimchon|
|590|Seth|seth.hannon@sakil...|al-Manama|
|8|Susan|susan.wilson@saki...|Hamilton|
|78|Lori|lori.wood@sakilac...|Rae Bareli|
|232|Constance|constance.reid@sa...|Zaria|
|12|Nancy|nancy.thomas@saki...|Yamuna Nagar|
```

474	Dustin dustin.gillette@s...	Emmen
191	Jeanette jeanette.greene@s...	Novoterkassk
230	Joy joy.george@sakila...	Botosani
511	Chester chester.benner@sa...	Suihua
493	Brent brent.harkins@sak...	Sultanbeyli
329	Frank frank.waggoner@sa...	Jamalpur
407	Dale dale.ratcliff@sak...	Rustenburg
595	Terrence terrence.gunderso...	Jinzhou
140	Eva eva.ramos@sakilac...	Clarksville
496	Tyler tyler.wren@sakila...	Rizhao
350	Juan juan.fraley@sakil...	Teboksary
485	Clyde clyde.tobias@saki...	Shaoguan
181	Ana ana.bradley@sakil...	Memphis
441	Mario mario.cheatham@sa...	Batna
239	Minnie minnie.romero@sak...	Ciomas
505	Rafael rafael.abney@saki...	Talavera
6	Jennifer jennifer.davis@sa...	Laredo
554	Dwayne dwayne.olvera@sak...	Rajkot
541	Darren darren.windham@sa...	Livorno
524	Jared jared.ely@sakilac...	Purwakarta
158	Veronica veronica.stone@sa...	Greensboro
423	Alfred alfred.casillas@s...	Sawhaj
148	Eleanor eleanor.hunt@saki...	Saint-Denis
122	Thelma thelma.murray@sak...	Lima
540	Tyrone tyrone.asher@saki...	Ruse
486	Glen glen.talbert@saki...	Acua
101	Peggy peggy.myers@sakil...	Abha
413	Marvin marvin.yee@sakila...	Berhampore (Bahar...
431	Joel joel.francisco@sa...	Sucre
100	Robin robin.hayes@sakil...	Jelets
90	Ruby ruby.washington@s...	Meixian
389	Alan alan.kahn@sakilac...	Emeishan
235	Jackie jackie.lynch@saki...	Buenaventura
313	Donald donald.mahon@saki...	Ezhou
88	Bonnie bonnie.hughes@sak...	Cheju
223	Melinda melinda.fernandez...	Mandi Bahauddin
163	Cathy cathy.spencer@sak...	Kakamigahara
202	Carla carla.gutierrez@s...	Bhusawal
304	David david.royal@sakil...	Callao
167	Sally sally.pierce@saki...	Chandrapur
171	Dolores dolores.wagner@sa...	Ipoh
518	Gabriel gabriel.harder@sa...	Sivas
331	Eric eric.robert@sakil...	Santa F
432	Edwin edwin.burk@sakila...	Newcastle
126	Ellen ellen.simpson@sak...	Po
147	Joanne joanne.robertson@...	Urawa
314	George george.linton@sak...	Sorocaba
404	Stanley stanley.scroggins...	Omiya

468	Tim tim.cary@sakilacu...	Bijapur
273	Priscilla priscilla.lowe@sa...	Salamanca
359	Willie willie.markham@sa...	Almirante Brown
582	Andy andy.vanhorn@saki...	Huejutla de Reyes
307	Joseph joseph.joy@sakila...	Angra dos Reis
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